

Research Idea: NGC2492

ALMA Thesis Planner (automated pipeline)

Title: A uniform, completeness-calibrated census of millimeter beam flux at the nuclei of "large photon ring" SMBHs: the >8 vs >2 microarcsec detection-rate contrast in ALMA project 2023.1.01099.S

The question the thesis stands on. These SMBHs were selected to be faint at centimeter wavelengths with weak or absent jets, and the proposal states plainly that their flux in the ~ 230 GHz band the project targets is unknown. The thesis measures it uniformly and asks one primary scientific question: **does the detection rate of millimeter emission differ between the >8 microarcsec "large ring" tier and the >2 microarcsec piggyback tier, and between SMBHB candidates and single SMBHs?** Those two splits are the axes the project itself used to build its target list, both are given as labels in the project description, and neither has ever been measured. The secondary question is the survey's operational one: what fraction of candidates clear the ~ 10 mJy nuclear-flux level the project uses for 12 m / VLBA promotion, and the "few 10s of mJy" level quoted as the passive-phased EHT detection floor.

What is actually measured, stated once and carried through everything. The ACA cannot separate a nucleus from its immediate circumnuclear surroundings. The measured quantity is therefore **beam flux density at the catalog nuclear position** — a well-defined observable with genuine detections and genuine non-detections. Every statistical statement in the thesis is a statement about the *beam flux* distribution. That distribution then bounds the nuclear one in a single, explicit direction: nuclear flux $\leq$ beam flux, so the measured rate above any threshold is an upper bound on the true nuclear rate. One inequality, stated in the abstract, applied at the end — not a second set of "corrected" numbers derived from an assumed subtraction the data cannot constrain. What the bound integrates over is quantified instead: for every field, the synthesized beam and the maximum recoverable scale are tabulated, so the reader knows the angular scale (and, where PI-supplied host distances allow, the physical scale) inside which emission is unresolved and the scale above which it would have been resolved out.

Frequency provenance. The data description says "230 GHz" without stating whether that figure is rest-frame or observed-frame; it is treated here as a nominal Band 6 label only, and the thesis says so explicitly rather than assuming either framing. Per-target observed sky frequencies, bandwidths, and synthesized beams are read from the SPW and image headers and reported as the effective frequency of each measurement.

Week-one gating steps, before sign-off.

- *Archive query.* Resolve the proprietary status of 2023.1.01099.S field by field and record the exact number of released fields, so working sample size is a known quantity rather than a risk.
- *Metadata, as bonus not dependency.* Request predicted ring diameters and per-target SMBHB

flags from the PI team. These are treated as a *bonus covariate* enabling a continuous analysis; the primary analysis needs only the tier labels the project description already supplies.

- *Disk audit.* Pipeline continuum images are cheap; calibrated MSes for the uv-plane cross-check are not. Budget $\sim 5\text{--}15$ GB per calibrated ACA Band 6 dataset, channel-average to continuum with ‘split’ immediately on retrieval, and stage in batches of ~ 25 , keeping the working set at a few hundred GB.

Pre-registered sample (fixed before any flux is examined), ~ 120 released fields committed. Tier A: all released targets from the 68 with predicted ring > 8 microarcsec. Tier B: a random draw, seed written into the thesis proposal, from the released > 2 microarcsec piggyback fields and the 124 SMBHB candidates, sized to bring the total to ~ 120 and to balance the two comparison cells. The rule is defined entirely on tier labels, so it is executable regardless of what metadata arrives. Fields beyond ~ 120 are explicit stretch goals, added only in the final semester if throughput allows, and their addition is reported as such. Fields failing QA are reported, not silently replaced. Explicitly not all 803.

Method. 1. Retrieve and stage calibrated MSes and pipeline continuum products per the audit. 2. One scripted, uniform measurement per field: forced photometry / image-plane fitting at the catalog nuclear position, cross-checked against a uv-plane point-source fit. Every field yields either a beam flux density with a full uncertainty budget (thermal plus flux-calibration systematic) or an upper limit at a stated confidence level, computed identically across the sample. Uniformity is the deliverable; no bespoke reductions. No compact/extended decomposition is attempted — the ACA’s $\sim 9\text{--}45$ m baselines give a uv lever arm of roughly a factor of 5 and cannot support one. 3. Completeness, analytically with an empirical check. Because photometry is forced at a known position, the completeness curve is a function of flux/rms alone and is written down analytically. It is then validated by full injection–recovery on a subsample of $\sim 15\text{--}20$ representative fields spanning the range of noise, beam size, and residual-map structure; the same exercise measures flux-recovery bias and scatter. If the subsample confirms the analytic form, it is used sample-wide; if it does not, the discrepancy is characterized and the injections are extended only as far as needed. This keeps the compute bounded to hundreds of ‘tclean’ calls, not tens of thousands. 4. Censored statistics on a single, consistent quantity. With the completeness curves in hand, fit the **beam flux** distribution using survival-analysis methods (Kaplan–Meier / censored maximum likelihood), so that real detections and real upper limits are both used correctly. Report the fraction above ~ 10 mJy and above the few-tens-of-mJy level with confidence intervals, then apply the nuclear-flux inequality once, in words, to the headline number. 5. The primary comparison. Test the Tier A versus Tier B detection-rate contrast, and the SMBHB-candidate versus single contrast, using two-sample censored tests (e.g. logrank on the censored beam-flux distributions). Power for these two comparisons is computed from the analytic completeness curves *before* observing the answer, and the pre-computed minimum detectable rate difference is stated in the proposal. If the data cannot separate the cells at the achieved sample size, that is the reported result, with the quantified upper limit on any difference.

Deliverables. (a) A machine-readable table of nominal-Band-6 nuclear-position beam flux densities and upper limits for ~ 120 fields, with per-target observed frequencies, beams, and resolved-out angular scales. (b) Analytic completeness curves plus their injection–recovery validation on the subsample. (c) The censored beam-flux distribution and detection rates above the ~ 10 mJy and few-tens-of-mJy thresholds, with the nuclear-flux bound stated. (d) The Tier A vs Tier B and SMBHB vs single contrasts with confidence intervals and pre-computed power. (e) The reusable pipeline, ready to run on the remaining fields.

Why it is tractable. Archival data, standard CASA tooling, one pipeline written once and run uniformly, no new observations, no external catalogs, no reliance on angular resolution the ACA does not have, no assumed-subtraction numbers, and one measured quantity carried consistently from photometry through the statistics. The answer is informative in every direction: if large-ring candidates are detected more often than the piggyback tier, the project’s selection strategy is validated as a way of finding EHT-feasible targets; if the rates are indistinguishable, ring size is not predictive of millimeter brightness and the survey’s triage should be rebuilt on other grounds; and if rates are low across the board, that is the strongest available statement that millimeter accretion emission in weak-jet SMBHs sits below EHT reach.